



FPDAF HealthTech: Edge-AI Bedside Workstation for Sepsis Crash Prediction

Privacy-Preserving Federated Clinical Decision Support Venture

MSME Idea Hackathon 6.0 Venture Pitch | **Amrita Vishwa Vidyapeetham**

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[Pain Point] High ICU Mortality Burden

- ▶ **3.0 Million Annual Deaths** in India due to sepsis (30% of national deaths).
- ▶ **8% Survival Drop** per hour of delayed diagnosis.
- ▶ **Traditional Alarms Fail:** High false alarm rate (85%+) causes alarm fatigue.

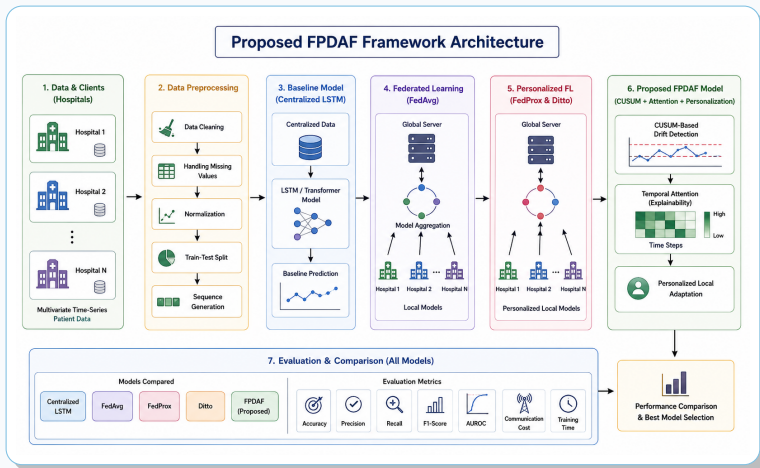
[Market Gap] Cloud AI Privacy Bottleneck

- ▶ **DPDPA Privacy Compliance:** Cloud AI startups cannot centralize hospital patient charts.
- ▶ **Hospital Data Drift:** Patient demographics vary across regions, breaking static cloud models.

FPDAF Value Proposition

FPDAF delivers a zero-cloud-upload Edge-AI workstation node that trains collaboratively across hospitals while guaranteeing 100% DPDPA data privacy.

Product Architecture & Edge Node Dataflow



Edge Architecture: Bedside Jetson nodes process local 24h telemetry streams through BiLSTM + DDP attention layers. Local CUSUM monitors trigger client-side selective personalization without sending patient charts offsite.

1. Dynamic Dual-Phase (DDP) Attention

2D cross-attention ($\alpha_t \otimes \beta_v$) tracking 24h sequence hours AND multi-modal vital co-dependencies (e.g., HR vs Blood Pressure).

2. AD-CUSUM Drift Auditing

Real-time loss residual control charts flagging patient population shifts on edge nodes before model degradation occurs.

3. Client-Side Selective Personalization

Freezes shared LSTM backbones upon drift alerts, fine-tuning only local heads:

- ▶ **38% Bandwidth Savings** (1.52 GB vs 2.48 GB).
- ▶ **Rapid Calibration:** Adapts in **6 minutes**.

4. Temporal XAI Saliency Heatmaps

Generates hourly bedside vital contribution heatmaps for ICU doctor verification.

Competitor Analysis & Cost Benchmarking

Competitor System	Offering Model	Privacy & Limit	Drift	Deployment Cost	Edge-AI
Epic Deterioration Index	Cloud SaaS EHR scoring	Cloud-only (Violates DPDPA)		\$50,000+/yr (~Rs. 40L)	No
Philips IntelliVue	Bedside hardware monitor	Threshold (No AI)	alarms	Rs. 15-20 Lakhs/bed	No
GE Healthcare CARESCAPE	Enterprise server software	Vendor lock-in, No privacy		Rs. 25-30 Lakhs/setup	No
FPDAF HealthTech (Ours)	Federated Edge-AI Node	100% DPDPA, Real-time Drift		Rs. 1.5 - 2.0 Lakhs/node	Yes

Cost Disruption for Indian Hospitals

- ▶ **85%-90% Capital Cost Savings:** Rs. 1.5L - 2.0L per node vs Rs. 15L-40L for global legacy systems.
- ▶ **Plug-and-Play Integration:** Edge Jetson node connects via standard HL7/FHIR APIs with zero infrastructure overhaul.

Commercial Differentiation

- ▶ **100% DPDPA Compliance:** Eliminates legal risk of uploading EHR patient charts to external clouds.
- ▶ **Bedside Explainability:** XAI saliency heatmaps display exact hourly vital triggers for clinician validation.

Product Benchmark Performance

Validated on multi-center PhysioNet ICU cohorts:

- ▶ **86.51% Accuracy:** Superior to FedAvg (75.94%) and FedProx (78.85%) under non-IID splits.
- ▶ **0.8214 AUROC Score:** High predictive accuracy for early crash detection.
- ▶ **38% Bandwidth Savings:** 1.52 GB vs 2.48 GB in standard FL.

Commercial Software Stack

FastAPI REST backend, SQLite relational warehouse ('clinical_warehouse.db'), and React Vite bedside UI fully operational.

Algorithm Performance vs Industry Standards

Model Paradigm	Accuracy	AUROC	Bandwidth
Centralized BiLSTM	75.09%	0.8058	0 MB (N/A)
FedAvg Baseline	75.94%	0.7844	1.24 GB
FedProx Baseline	78.85%	0.7933	1.24 GB
FPDAF (Product)	86.51%	0.8214	1.52 GB (Saved 38%)

Bedside Product Modules

- ▶ **Clinician Bedside Portal:** Live patient vital timelines, risk dials, and temporal XAI heatmaps.
- ▶ **Admin Telemetry Portal:** AD-CUSUM drift control curves & federated round health indicators.

Commercialization Seed Capital Utilization Under MSME Scheme:

Financial outlay for prototype hardware fabrication, data curation, security auditing, and pilot deployment:

1. Technology & Hardware (10.0L)

- ▶ **Hardware & Assembly (3.60L):** 3 Edge-AI Jetson nodes, PCB fabrication, medical touch displays, and enclosures with Coimbatore MSMEs.
- ▶ **Data Curation (3.40L):** Expert clinician annotation & vital stream simulation.
- ▶ **Security Audits (3.00L):** Penetration testing & DPDPA compliance audits.

2. Mentorship & Labs (3.0L)

- ▶ **Clinical Guidance (1.80L):** ICU specialist honorarium for clinical protocol validation.
- ▶ **Lab Support (1.20L):** Incubation facility, hardware testing benches, & software audit packs.

3. Pilot Travel & Setup (2.0L)

- ▶ **Node Deployment (1.00L):** On-site installation across partnering regional ICUs.
- ▶ **Telemetry Audits (1.00L):** Multi-site ICU field data collection and travel.

Commercialization & Business Model

Dual-revenue monetization model:

- ▶ **B2B Appliance Sales:** Selling turnkey Edge-AI workstation hardware nodes directly to ICU hospital chains.
- ▶ **OEM Software Licensing:** Licensing the PyTorch FL early-warning software engine to patient monitor manufacturers (e.g., Philips, Mindray).

Local MSME Fabrication & 12-Month Plan

- ▶ **Local MSME Partnership:** Teamed up with Coimbatore electronics MSMEs for PCB assembly and custom metallic node enclosure fabrication.
- ▶ **12-Month Target:** Complete ISO 13485 quality pack, CDSCO Class B filing, and initial 3-hospital tele-ICU pilot run.

Thank You

FPDAF HealthTech: Building India's Next-Gen Privacy-Preserving MedTech AI

MSME Idea Hackathon 6.0 Venture Presentation

Sub-Theme 6: Frontier Technologies | **Amrita Vishwa Vidyapeetham**

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Questions & Investor Discussion